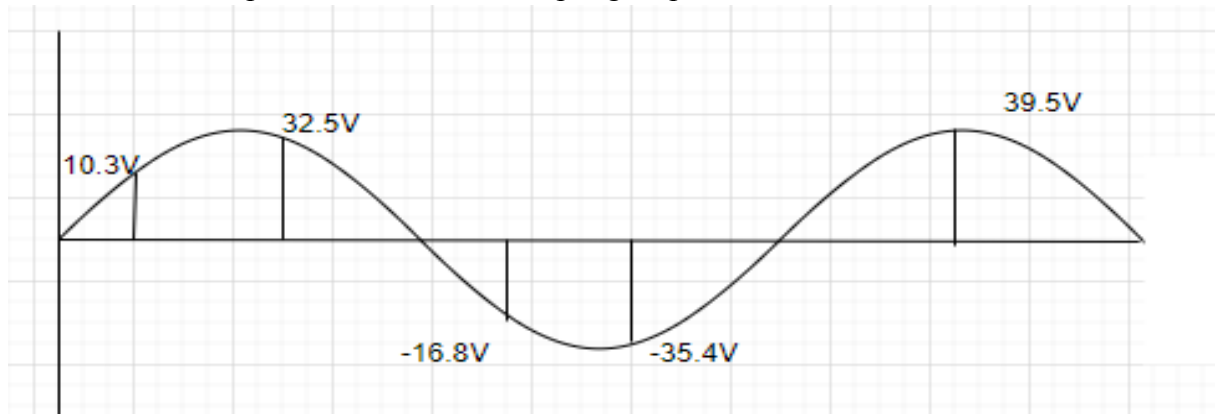


## PCM Solved Example

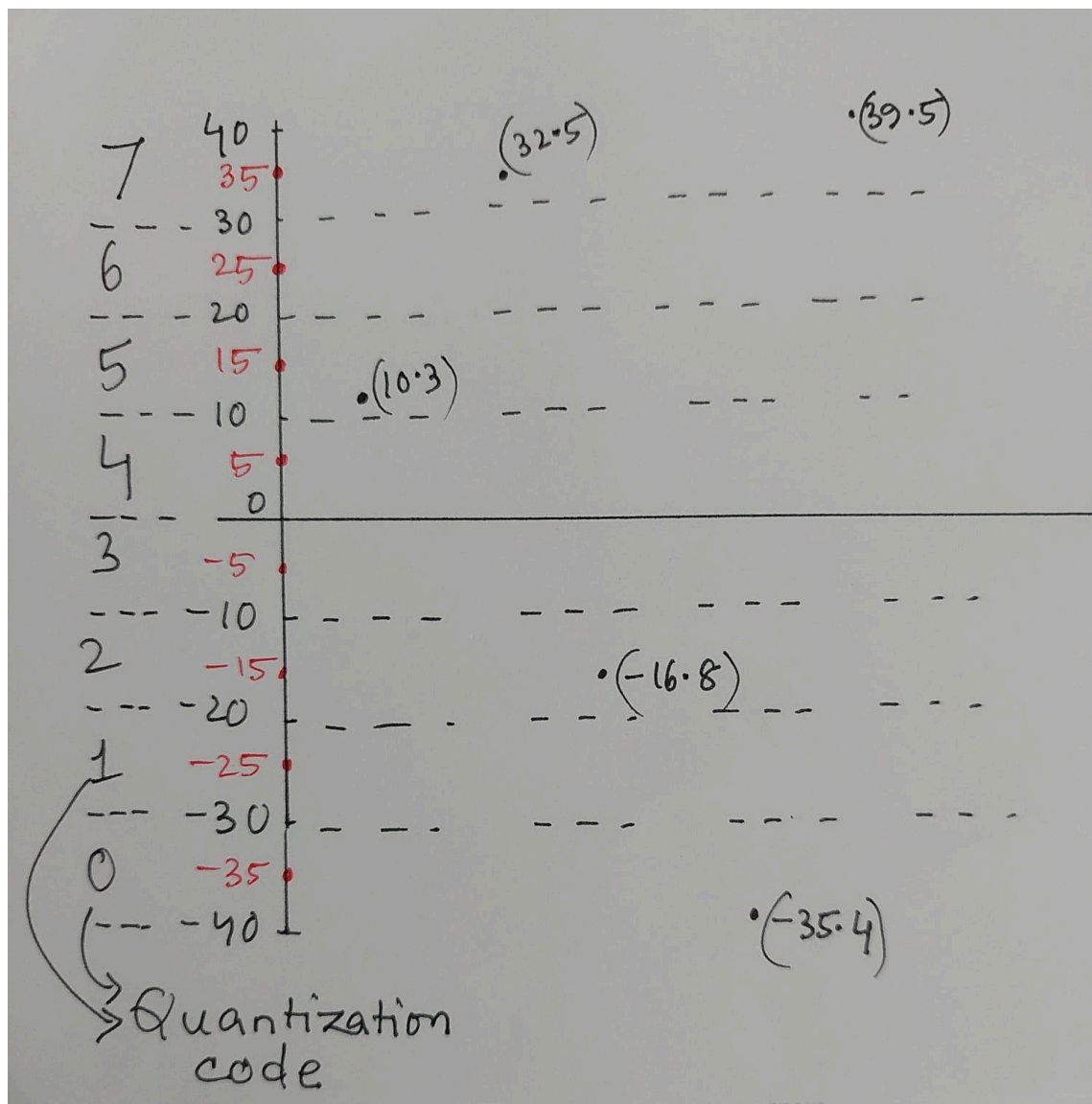
The following figure shows a sampled analog signal. Convert it to a digital signal using **Pulse Code Modulation**. Assume there will be **3-bit** code words for each sampled amplitude. **Show** the **normalized Quantized** value and **quantization code** for the given analog signal value at different time stamps. Assume that the sampling amplitudes are between -40V to +40V.



[Practice this yourself, then match with the solution in the next page. Just going through the solution will not be good for preparation.]

## Solution

|                                                             |     |
|-------------------------------------------------------------|-----|
| Lowest Amplitude = $V_{\min}$                               | -40 |
| Highest Amplitude = $V_{\max}$                              | 40  |
| Number of Levels = $L$                                      | 8   |
| Level Height (Delta) = $\Delta$                             | 10  |
| Number of Bits in One Level = $\log_2(\text{Levels}) = n_b$ | 3   |



|                                                                                     |      |      |       |       |       |
|-------------------------------------------------------------------------------------|------|------|-------|-------|-------|
| <b>Actual Amplitude = PAM value</b>                                                 | 10.3 | 32.5 | -16.8 | -35.4 | 39.5  |
| <b>Normalized PAM Values = (Actual Amplitude / Level Height)</b>                    | 1.03 | 3.25 | -1.68 | -3.54 | 3.95  |
| <b>Quantization Value = Approximate Adjacent Middle Value of Quantization Level</b> | 15   | 35   | -15   | -35   | 35    |
| <b>Normalized Quantized Values = (Quantization Value / Level Height)</b>            | 1.5  | 3.5  | -1.5  | -3.5  | 3.5   |
| <b>Normalized Error = Normalized Quantized Values - Normalized PAM Values</b>       | 0.47 | 0.25 | 0.18  | 0.04  | -0.45 |
| <b>Quantization Code</b>                                                            | 5    | 7    | 2     | 0     | 7     |
| <b>Encoded Bit</b>                                                                  | 101  | 111  | 010   | 000   | 111   |

**Final Digital Signal: 101111010000111**